

```
In [3]: import pandas as pd
import numpy as np
import matplotlib
```

```
In [ ]: #RAW DATA EXTRACTION
```

```
In [4]: df = pd.read_csv('online_retail_II_reduced.csv')
```

```
In [5]: df
```

Out[5]:

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	01-12-2009 07:45	6.95	13085.0	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	01-12-2009 07:45	6.75	13085.0	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	01-12-2009 07:45	6.75	13085.0	United Kingdom
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	01-12-2009 07:45	2.10	13085.0	United Kingdom
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	01-12-2009 07:45	1.25	13085.0	United Kingdom
...	...	...	...	...	...	...	...	...
11571	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	04-12-2009 14:56	5.95	17151.0	United Kingdom
11572	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	04-12-2009 14:56	5.45	17151.0	United Kingdom
11573	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	04-12-2009 14:56	5.45	17151.0	United Kingdom
11574	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	04-12-2009 14:56	5.45	17151.0	United Kingdom
11575	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	04-12-2009 15:11	1.25	16451.0	United Kingdom

11576 rows × 8 columns

```
In [ ]: #RENAMING COLUMNS TO READABLE FORM
```

```
In [6]: df.rename(columns={'Customer ID': 'Customer_ID'})
```

```
Out[6]:
```

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer_ID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	01-12-2009 07:45	6.95	13085.0	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	01-12-2009 07:45	6.75	13085.0	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	01-12-2009 07:45	6.75	13085.0	United Kingdom
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	01-12-2009 07:45	2.10	13085.0	United Kingdom
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	01-12-2009 07:45	1.25	13085.0	United Kingdom
...	...	...	...	...	...	...	...	...
11571	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	04-12-2009 14:56	5.95	17151.0	United Kingdom
11572	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	04-12-2009 14:56	5.45	17151.0	United Kingdom
11573	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	04-12-2009 14:56	5.45	17151.0	United Kingdom
11574	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	04-12-2009 14:56	5.45	17151.0	United Kingdom
11575	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	04-12-2009 15:11	1.25	16451.0	United Kingdom

11576 rows × 8 columns

```
In [5]: #CHECKING NULL VALUES
```

```
In [7]: df.isnull().any()
```

```
Out[7]: Invoice      False
StockCode    False
Description    True
Quantity      False
InvoiceDate   False
Price         False
Customer ID    True
Country       False
dtype: bool
```

```
In [ ]: #Removed Customer ID with null Values
```

```
In [8]: df = df.dropna(subset=['Customer ID'])
df
```

Out[8]:

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	01-12-2009 07:45	6.95	13085.0	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	01-12-2009 07:45	6.75	13085.0	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	01-12-2009 07:45	6.75	13085.0	United Kingdom
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	01-12-2009 07:45	2.10	13085.0	United Kingdom
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	01-12-2009 07:45	1.25	13085.0	United Kingdom
...	...	...	...	...	...	...	...	...
11571	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	04-12-2009 14:56	5.95	17151.0	United Kingdom
11572	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	04-12-2009 14:56	5.45	17151.0	United Kingdom
11573	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	04-12-2009 14:56	5.45	17151.0	United Kingdom
11574	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	04-12-2009 14:56	5.45	17151.0	United Kingdom
11575	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	04-12-2009 15:11	1.25	16451.0	United Kingdom

8528 rows × 8 columns

In [9]:

```
df.isnull().any()
```

Out[9]:

Invoice	False
StockCode	False
Description	False
Quantity	False
InvoiceDate	False
Price	False
Customer ID	False
Country	False
dtype:	bool

In []:

```
#REMOVING NEGATIVE VALUES
```

In [10]:

```
df = df[df['Quantity']>0]      #Negative Quantity Removed
df
```

Out[10]:

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	01-12-2009 07:45	6.95	13085.0	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	01-12-2009 07:45	6.75	13085.0	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	01-12-2009 07:45	6.75	13085.0	United Kingdom
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	01-12-2009 07:45	2.10	13085.0	United Kingdom
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	01-12-2009 07:45	1.25	13085.0	United Kingdom
...	...	...	...	...	...	...	...	...
11571	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	04-12-2009 14:56	5.95	17151.0	United Kingdom
11572	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	04-12-2009 14:56	5.45	17151.0	United Kingdom
11573	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	04-12-2009 14:56	5.45	17151.0	United Kingdom
11574	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	04-12-2009 14:56	5.45	17151.0	United Kingdom
11575	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	04-12-2009 15:11	1.25	16451.0	United Kingdom

8342 rows × 8 columns

In [11]:

```
df = df[df['Price']>0] #Negative Price Removed
df
```

Out[11]:

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	01-12-2009 07:45	6.95	13085.0	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	01-12-2009 07:45	6.75	13085.0	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	01-12-2009 07:45	6.75	13085.0	United Kingdom
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	01-12-2009 07:45	2.10	13085.0	United Kingdom
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	01-12-2009 07:45	1.25	13085.0	United Kingdom
...	...	...	...	...	...	...	...	...
11571	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	04-12-2009 14:56	5.95	17151.0	United Kingdom
11572	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	04-12-2009 14:56	5.45	17151.0	United Kingdom
11573	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	04-12-2009 14:56	5.45	17151.0	United Kingdom
11574	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	04-12-2009 14:56	5.45	17151.0	United Kingdom
11575	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	04-12-2009 15:11	1.25	16451.0	United Kingdom

8340 rows × 8 columns

In [12]: *#CHANGING FORMATS*

In [13]: `df.loc[:, 'InvoiceDate']=pd.to_datetime(df['InvoiceDate']) #Converted InvoiceDate to date fromat.`
`df`

Out[13]:

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-01-12 07:45:00	6.95	13085.0	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085.0	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085.0	United Kingdom
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-01-12 07:45:00	2.10	13085.0	United Kingdom
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-01-12 07:45:00	1.25	13085.0	United Kingdom
...	...	...	...	...	...	...	...	...
11571	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	2009-04-12 14:56:00	5.95	17151.0	United Kingdom
11572	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	2009-04-12 14:56:00	5.45	17151.0	United Kingdom
11573	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	2009-04-12 14:56:00	5.45	17151.0	United Kingdom
11574	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	2009-04-12 14:56:00	5.45	17151.0	United Kingdom
11575	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	2009-04-12 15:11:00	1.25	16451.0	United Kingdom

8340 rows × 8 columns

In [14]: *#AGAIN CHANGING TO ORIGINAL DATAFRAME*

In [15]: `df=df.copy()`
df

Out[15]:

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-01-12 07:45:00	6.95	13085.0	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085.0	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085.0	United Kingdom
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-01-12 07:45:00	2.10	13085.0	United Kingdom
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-01-12 07:45:00	1.25	13085.0	United Kingdom
...	...	...	...	...	...	...	...	...
11571	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	2009-04-12 14:56:00	5.95	17151.0	United Kingdom
11572	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	2009-04-12 14:56:00	5.45	17151.0	United Kingdom
11573	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	2009-04-12 14:56:00	5.45	17151.0	United Kingdom
11574	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	2009-04-12 14:56:00	5.45	17151.0	United Kingdom
11575	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	2009-04-12 15:11:00	1.25	16451.0	United Kingdom

8340 rows × 8 columns

```
In [ ]: #CREATING A MONETARY VALUE FOR RFM ANALYSIS
```

```
In [16]: df['TotalAmount']=df['Quantity'] * df['Price'] #Created monetary value of the product for RFM analysis
df
```

Out[16]:

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country	TotalAmount
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-01-12 07:45:00	6.95	13085.0	United Kingdom	83.4
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085.0	United Kingdom	81.0
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085.0	United Kingdom	81.0
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-01-12 07:45:00	2.10	13085.0	United Kingdom	100.8
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-01-12 07:45:00	1.25	13085.0	United Kingdom	30.0
...	...	...	...	...	...	...	...	...	...
11571	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	2009-04-12 14:56:00	5.95	17151.0	United Kingdom	35.7
11572	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	2009-04-12 14:56:00	5.45	17151.0	United Kingdom	65.4
11573	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	2009-04-12 14:56:00	5.45	17151.0	United Kingdom	65.4
11574	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	2009-04-12 14:56:00	5.45	17151.0	United Kingdom	65.4
11575	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	2009-04-12 15:11:00	1.25	16451.0	United Kingdom	15.0

8340 rows × 9 columns

```
In [ ]: #FINAL DATA TYPE CHECK
```

```
In [17]: df.dtypes
```

Out[17]: Invoice object
StockCode object
Description object
Quantity int64
InvoiceDate object
Price float64
Customer ID float64
Country object
TotalAmount float64
dtype: object

```
In [19]: print("Shape of the data", df.shape)  
df
```

Shape of the data (8340, 9)

Out[19]:

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country	TotalAmount
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-01-12 07:45:00	6.95	13085.0	United Kingdom	83.4
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085.0	United Kingdom	81.0
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-01-12 07:45:00	6.75	13085.0	United Kingdom	81.0
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-01-12 07:45:00	2.10	13085.0	United Kingdom	100.8
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-01-12 07:45:00	1.25	13085.0	United Kingdom	30.0
...	...	...	...	...	...	...	...	...	...
11571	490310	21551	CERAMIC BIRDHOUSE BUTTERFLY LARGE	6	2009-04-12 14:56:00	5.95	17151.0	United Kingdom	35.7
11572	490310	21550	CERAMIC BIRDHOUSE CRESTED TIT LARGE	12	2009-04-12 14:56:00	5.45	17151.0	United Kingdom	65.4
11573	490310	21552	CERAMIC BIRDHOUSE BLACK ROSE LARGE	12	2009-04-12 14:56:00	5.45	17151.0	United Kingdom	65.4
11574	490310	21554	CERAMIC BIRDHOUSE RED ROOF LARGE	12	2009-04-12 14:56:00	5.45	17151.0	United Kingdom	65.4
11575	490311	21232	STRAWBERRY CERAMIC TRINKET BOX	12	2009-04-12 15:11:00	1.25	16451.0	United Kingdom	15.0

8340 rows × 9 columns

```
In [ ]: #EXPORTING CLEANED DATA

In [ ]: #df.to_excel("cleaned_data_phase_1.xlsx", index=False) #Imported the file to xlsx format
```